

Research Article

Psychological restoration, place attachment and satisfaction in birders and non-birding visitors[☆]Christoph Randler^{*}, Selina Friedrich, Sophia Koch (née Nagel)

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ABSTRACT

Travelling to special places to observe birds is becoming an important niche in ecotourism. Here, we addressed differences between birdwatchers and non-birders to gain insight into the different needs (satisfaction, psychological restoration, and place attachment) at the same locality and during the identical time period at eight birdwatching sites in SW Germany. The study took place between April 9, 2022 and June 26, 2022; N = 426 participants were surveyed with a questionnaire on site. Non-birders had a higher visit satisfaction, while birders had a higher loyalty, had a longer visit duration, and higher visitation frequency. Visit satisfaction was positively correlated with place attachment and psychological restoration. Destination loyalty was unrelated to psychological restoration in birders but was positively related in non-birders. Experiencing a subjective highlight, like a specific bird species or encountering an interesting animal, led to higher psychological restoration and visit satisfaction in both groups, but non-birders generally had a higher psychological restoration and visit satisfaction. Place attachment was not predicted by a highlight nor by being a birder or not.

Management implications: Visitors of bird rich areas or nature reserves differ strongly whether they are birders or not. As birders invest more effort in their visit (travel, revisit, loyalty) they expect more from their visit (e.g., Stemmer et al., 2022). We also emphasized the importance of a highlight, which makes a visit for non-birders more attractive. Highlights, for example, are observations of a bird that have a personal meaning, and that may be fascinating or just interesting to observe. Therefore, it is important that managers of parks or nature reserves draw specific attention to highlights that are currently observable and that increase visitor satisfaction. This can be done by providing information (Hughes & Morrison-Saunders, 2002), such as pop-up presentations or by announcing it on webpages. However, highlights are different among birders and non-birders and this needs to be taken into account, e.g., by flexible announcements. To 'hear' what the tourists say (Prakash et al., 2019) is an important management tool to tailor visitor experiences, and birders and non-birders differ, so that nature reserves should offer different strains of experience for both, the specialized birder and the nature-interested tourist with lower knowledge.

1. Introduction

Outdoor recreation should take social, situational, and ecological factors into account (Ferguson et al., 2022) to study visitor experience and satisfaction during nature experience. One aspect of outdoor recreation and nature experience – wildlife watching – has become an increasingly important branch of tourism (Lucrezi et al., 2022; Mutanga et al., 2017). Especially avitourism, i.e., travelling to special places to observe birds, is gaining increasing importance as a niche in ecotourism (Callaghan et al., 2018; Hvenegaard et al., 1989). However, only few

studies assessed the differences between birdwatchers and non-birders to gain insight into the different needs and attitudes at the same locality and during the identical time period (Hvenegaard & Dearden, 1998). As tourist heterogeneity has been previously emphasized as important research goal (Mutanga et al., 2017), birders and non-birders may be different in their needs and expectations when visiting a birdwatching site (Kumar et al., 2020). Therefore, we compare birders and non-birders at eight birding hotspots to study satisfaction, psychological restoration and place attachment in birders and non-birders. We especially focus on the experience of a subjective highlight during the visit, because

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previous research showed the importance of wildlife experience and nature observation related to visitor satisfaction (Kuldna et al., 2020; Mutanga et al., 2017). As place attachment (e.g., Beery & Jönsson, 2017; Kil et al., 2021) and psychological restoration have been studied widely in tourism, we used these outcome variables in addition to visitor satisfaction for the comparison of both groups, birders and non-birders. This study should help gaining insight and understanding the differences between birders and non-birders in psychological restoration, satisfaction, and place attachment.

1.1. Ecotourism

Ecotourism is one of the fastest growing segments in tourism (Chen & Chen, 2015; Dowling & Stephen, 2002). Ecotourism is a type of sustainable tourism that focuses on preserving and conserving the natural environment and its wildlife, while also providing educational and cultural opportunities for visitors (Goodwin, 1996). One objective of ecotourism is to minimize negative impacts on the environment and promote conservation efforts, while allowing visitors to learn about and experience the natural world (Tsaour et al., 2006; Wood, 2002). A subcategory or niche of ecotourism is avitourism, also known as birding tourism, which specifically focuses on birdwatching and the observation of wild bird species in their natural habitats (Steven et al., 2015). Avitourism destinations may include national parks, wildlife reserves, wetlands, and other areas that are important habitats for bird species. The goal of avitourism is to provide visitors with opportunities to experience the beauty and diversity of bird species, while also supporting conservation efforts and promoting sustainable tourism practices (Steven et al., 2015). Avitourism involves both, guided tours and self-organized independent birdwatching. Birdwatching or birding itself is defined as the act of observing and identifying birds in their native habitats (Sekercioglu, 2002). Birdwatching, in turn, is referred to as 'avitourism' or 'birding tourism' if the birder undertakes a trip of a mile or more from home for the primary purpose of observing birds in a natural setting (Conradie, 2015).

To quantify the impact of avitourism, different studies have been carried out (Kruger & Viljoen, 2023; Kumar et al., 2020; Steven et al., 2015). By reviewing 66 research studies published between 1989 and 2014 Steven et al. (2015) showed a distinct northern hemisphere bias, dominated by studies of the avitourists themselves, and mostly focusing on the economic impacts and birding motivations. Some studies were of bird-related events, e.g., festivals or challenges (Eubanks et al., 2004; Kumar et al., 2020; Scott et al., 1999; Scott & Thigpen, 2003; Steven et al., 2015). However, concerning psychological benefits or the comparison of birders and non-birders and their – possibly differing – needs have not been in the focus of earlier studies. Therefore, we focus on the three psychological outcomes, visitor satisfaction, psychological restoration and place attachment.

1.2. Visitor satisfaction and loyalty

Visitor satisfaction is a key concept in travel research (Cohen et al., 2014) and Ferguson et al. (2022) showed that social, situational and ecological factors should be integrated in studies on wildlife tourism. Heterogeneity of visitors in general, and differences between consumptive and non-consumptive visitors have been reported (Vaske & Roemer, 2013), with lower satisfaction scores in consumptive visitors. Although birding is a non-consumptive leisure activity, birders can be somewhat placed near the consumptive visitors, such as hunters or anglers (van den Heuvel et al., 2022), because birding can also yield harvest in terms of new bird species for a life-list, photographs or long lists of bird species observed (Randler, 2023; Randler & Großmann, 2022). Indeed, Goodfellow (2013) compared trophy hunting and birding and reported some similarities in avitourists, thus, one would expect lower satisfaction in birders compared to non-consumptive, non-birding visitors. Going into more detail, satisfaction with a specific element as an

antecedent of overall visit satisfaction was postulated by Dybsand et al. (2021). We here formalized this satisfaction with a specific element (*sensu* Dybsand et al., 2021) as a subjectively self-reported highlight of the visit. The subjectiveness is important, because birders and non-birders can differ immense in the definition of an avian "highlight" or charismatic species (Randler et al., 2023), with higher specialized birders often preferring small, dull, rare and shy species (Randler et al., 2023). Mutanga et al. (2017) emphasized that satisfaction with wildlife tourism was significantly predicted by a good wildlife experience. Thus, we assume that a wildlife experience (a highlight) should lead to a higher satisfaction, but also, different expectations of birders and non-birders are related to visitor satisfaction.

1.3. Place attachment

Place attachment is related to the "range of feelings humans associate with specific environments" (Kyle et al., 2005, p. 153), and thus addresses bonds between humans and specific places (Bryce et al., 2016; Kyle et al., 2005). These bonds can be variously described, for example as place attachment, sense of place, sense of community, place identity, place dependence or focus more on the therapeutic value of the place (Bryce et al., 2016; Hidalgo & Hernandez, 2001; Subiza-Pérez et al., 2021). Although recent research showed additional variables and dimensions for measuring place attachment, still place identity and place dependence seem to be key variables (Kyle et al., 2005; Lee, 2011; Williams & Vaske, 2003). Most studies in place attachment have focused on the neighborhood (Hidalgo & Hernandez, 2001), but different spatial scales have been studied (Lewicka, 2011), following concentric circles from neighborhood, home, city and, rarely, national regions (Lewicka, 2011). Many studies on place attachment have been carried out in the residential area of people, but another trend noted by Lewicka (2011) is the increasing focus on recreation places, such as parks and recreational areas (Kil et al., 2012; Yuan & Wu, 2021). For example, place attachment and well-being are positively correlated in outdoor recreational activities (Yuan & Wu, 2021) and place attachment is linked to restorative quality (Zhang et al., 2023). In addition to the extension to outdoor and leisure activities, place attachment has also gained relevance in tourism research (Dwyer et al., 2019). Place attachment may be different between birders and non-birders because birders may visit birdwatching sites because they feel related or attached to the place.

1.4. Psychological restoration

Direct contact with nature and biodiversity helps people to recover psychologically and has widely recognized positive effects on human health (Fisher et al., 2021; Kaplan & Kaplan, 1989; Ulrich et al., 1991; White et al., 2020). Two general theories are invoked to explain the relation between psychological well-being and nature. Attention Restoration Theory (ART) suggests, that spending time in nature allows recovering from attention fatigue (Kaplan, 1995; Kaplan & Kaplan, 1989). The stimuli provided by nature should be on an average level to allow attention to be restored. Stress Reduction Theory (SRT; Ulrich et al., 1991) postulates, that high levels of biodiversity benefit psychophysiological stress recovery. Following both theories, a stay in nature improves psychological health and well-being. As a stay in nature is in most cases related to leisure (because most of today's jobs are indoor), well-being and outdoor recreation are positively linked.

The positive relationship between place attachment and psychological restoration has been developed by Korpela and Hartig (1996) and Korpela et al. (2001). The restorative experience with a certain place can improve the identification of people with this place and strengthen attachment with that place (Korpela et al., 2001; Korpela & Hartig, 1996). Thus, in recent studies, psychological restoration and place attachment have been found to be positively correlated (Bazrafshan et al., 2023; Fleming et al., 2022; Liu et al., 2021; Subiza-Pérez et al., 2021). Further, both constructs of place attachment—place dependence

and place identity—positively predicted restorative potential (Liu et al., 2021; Ratcliffe & Korpela, 2016). Both groups, birders and non-birders should benefit from a visit to a natural area, but experiencing a highlight might lead to a higher psychological restoration.

1.5. Birders and non-birders

Birding has become an increasingly popular leisure activity (Rutter et al., 2021), but comparisons of different groups of tourists, e.g., with a strong bird interest versus non-birding ecotourist are scarce and dealt mainly with motivations and environmental beliefs (Hvenegaard, 2002; Hvenegaard & Dearden, 1998; Mutanga et al., 2017; Wurzinger & Johansson, 2006). Previously, ecotourists were compared with people carrying out other activities, at other locations, or at different time periods (Bimonte & Faralla, 2012; Hvenegaard & Dearden, 1998). When comparing groups at the same place, birders were found to have had higher travel costs compared to non-birders (Czajkowski et al., 2014), and even within birders willingness to pay was dependent on the motives, such as seeing special bird or aiming for a high quantity of species (Steven et al., 2017). Watching birds and experiencing nature may not only be positive for birders, but may also benefit many, if not all people (Tryjanowski et al., 2022). However, in the case of birders, more specialized birders were able to identify more bird species in an identification test (Randler & Heil, 2021). We therefore expect differences between psychological restoration, place attachment, and visit satisfaction between birders and non-birders because of their differing knowledge. In addition, other variables related to behavior, e.g., in trail use (Troped et al., 2009), that have previously used to characterized visits to nature and other areas were studied to gain an insight into the differences between birders and non-birders, but also to assess the relationship between those variables and our psychological outcomes, like visit frequency, duration travel distance, willingness to pay for a hypothetical entrance fee (Ribet & Brander, 2020).

Summarizing the literature review, we developed four main research questions.

RQ1. Do birders differ in their psychological restoration, place attachment and satisfaction from non-birders?

As shown above, there are differences in satisfaction among heterogeneous groups of nature visitors, and the birders differ from non-birders in many needs and expectations. Therefore, we used the outcome variables visitor satisfaction, place attachment and psychological restoration to analyze differences between both groups.

RQ2. Do birders differ from non-birders in variables that characterize their visit, such as willingness to pay and entrance fee, visit duration or travel distance? Similar to RQ2, we studied these differences because we assume that birders have a different motivation to visit a specific birdwatching site, and may travel longer, stay longer and may pay a higher hypothetical entrance fee.

RQ3. Are there differences concerning the associations (correlations) between visit characteristics, like travel distance, willingness to pay and entrance fee and visit duration on the one side and psychological restoration and place attachment on the other? As previous studies have shown a relationship between characteristics of a visit, we assume a correlation between these variables and place attachment and psychological restoration. However, as birders and non-birders differ, we look whether the correlations are in the same or an opposite direction.

RQ4. Does the experience of a subjective highlight impact differently on psychological restoration, satisfaction and place attachment of birders and non-birders? Based on the above-mentioned studies on experiencing wildlife and the importance of a “good wildlife experience” (Mutanga et al., 2017) we asked whether there exist differences in the outcome variables satisfaction, psychological restoration and place attachment between birders and non-birders with regard to

experiencing a highlight.

2. Materials and methods

2.1. Study places

The study was carried out in SW Germany, an area that hosts about 200–220 breeding bird species and some 100 migrant birds. It is a temperate climate with cool winters (average temperatures around freezing or slightly above), and warm summers with temperatures reaching the mid to high 20 s °C. The vegetation consists of a mix of deciduous and coniferous forests, along with agricultural areas in the flatter regions. The land use is characterized by a balance between urban areas and farmlands. We selected eight different places in SW Germany that all offer some basic commodities for birders, such as observation towers, or platforms to enhance birdwatching (Fig. 1). All places contained blue space, i.e., some freshwater (lake, pond) and are more lowland areas, no mountainous regions.

These places are all among the best birdwatching sites in this area but differ in their species richness. We used the eBird HotSpot function to check which places have a good number of bird species (checked on July 2, 2021) and the *orniho.de* website to gain an insight into the visitation frequency of birders at this place. Further, a common German book about birding places was used (Falke Redaktion, 2016). The study samples are shown according to the different places in Table 1. For every place, we extracted the number of species observed during the year 2022 from the eBird database (eBird data on August 2, 2022, from year 2022, all observations).

2.2. Participants, data collection and ethical considerations

The study took place between April 9, 2022 and June 26, 2022. A total of 426 participants filled the questionnaires collected at 8 different places. Demographic details for every place are depicted in Table 1. We used on-site surveys with questionnaires (Randler et al., 2007). The method was adapted from Dallimer et al. (2012). The surveys were carried out by SN & SF. They took between 10 and 15 min and were based on a standardized and consistent method of guiding visitors through the process. The participants filled the questionnaires on-site with a clip board and a pen. Prior to starting, each participant was given a brief project description and an assurance of anonymity. Written informed consent was obtained and the consent form was stored separately from the questionnaires. The participants answered the questions unobserved and secretly and put their questionnaire into an envelope and afterwards this envelope into an urn to guarantee anonymity. Participation was voluntary and the pen was given as compensation. The study was approved by the ethics committee of the faculty of economics and social sciences of the University of Tübingen (Az.: A2.5.4.-204_ns). The questions about the psychological state/restoration are widely used for this kind of research and the approval by the ethics committee, as well as the anonymized procedure leads to our assumption that the study was done in a most possible ethical manner.

2.3. Questionnaire

The questionnaire asked for the main reason for the visit, demographic variables (age, gender, university degree in a dichotomous manner). Then birders were asked for their birding specialization and non-birders were asked to continue overleaf.

2.4. Place attachment

We used three items to measure place attachment based on place identity (Boley et al., 2021). “I am very attached to this place.”, “This place is very special to me.”, and “I identify strongly with this place.” These items were five-point Likert scaled from “fully disagree” to “fully



Fig. 1. Overview over the birdwatching sites in SW Germany.

Table 1

Study characteristics from the eight different places (Table 1).

	Species	N	non-birder	birder	Mean age	SD age	with degree	no degree	% with degree
Altmühlsee	146	62	29	33	54.73	14.89	35	25	58%
Eriskircher Ried	85	34	26	8	58.39	13.94	22	8	73%
Federsee	103	116	59	57	53.4	14.52	36	75	32%
Pleidelsheim Wiesental	73	17	2	15	57.88	12.57	5	12	29%
Taubergießen	74	14	10	4	53.93	14.56	5	9	36%
Wagbachniederung	152	105	12	93	51.29	13.52	61	40	60%
Zugwiesen	86	60	41	19	50.27	14.39	39	20	66%
Freiburger Rieselfeld	74	18	7	11	42.22	16.34	12	6	67%
Total		426	186	240	52.73	14.52	215	195	52%

agree” Cronbach’s α of the sample was .866.

2.5. Psychological restoration

Psychological restoration was measured by four items, one depicted as smiley on a five-point scale, and three items on a five-point Likert

scale, from “fully disagree” to “fully agree”. The three-item measure for self-reported psychological restoration were adapted and slightly changed from Wyles et al. (2019), Nikunen and Korpela (2009), and Haugan (2015). The items were “I feel calm and relaxed,” “I feel refreshed”, and “I feel peaceful” and rated on a Likert scale from 1 (strongly disagree) to 5 (strongly agree). The Cronbach’s α was 0.765.

Mean scores were calculated for the scale.

2.6. Additional variables

Variables measured by single items were: Visitation frequency was scaled from 1 to 7 (How often do you visit this place? For the first time, about every year, a few times per year, about monthly, a few times per months, nearly weekly, few times per week). Travel distance was measured from 1 to 7 (How far did you travel? <10 km, 10–25 km, 26–50 km, 51–100 km, 101–250 km, 251–500 km, >500 km). Highlight of the visit (open question: Which bird or which experience was a highlight/something special for you today?). Overall visitor satisfaction was measured from 1 to 5 (“All in all, how satisfied were you with your visit today?” not at all satisfied, not really satisfied, partially satisfied, very satisfied, absolutely satisfied; e.g. following Czeszczewik et al., 2019). Loyalty was asked with a question about a revisit in the next year, coded 1 to 5 (“I will visit this area again next year.” From fully disagree to fully agree). Willingness to pay was measured by a hypothetical entrance fee (“If there was an entry fee here, how much would you be willing to pay?” 0 €, <10 €, 11–20 €, 21–30 €, 31–40 €, 41–50 €, >50 €). Visit duration was measured by (“Approximately how long did you stay in this area today?” < 1 h, 1–2 h, 2–3 h, 4–5 h, >5 h).

2.7. Statistical analysis

We used an effect size calculation to help estimate the number of respondents needed for the survey study, assuming a small to medium effect and four different groups related to our research questions, based on birders and non-birders and the experience of a highlight. The a priori effect size calculations were done with G*Power 3.1 (Faul et al., 2009). Based on an effect size f of 0.25, α error probability of 0.05, and power of 0.95, sample size was estimated 400 (with 10 numerator df, 4 groups (birder vs. non-birder, highlight vs. without highlight), one covariate). This suggests aiming for a sample size of $N = 400$ respondents. Non-parametric tests were used for single items because they were ordinaly scaled, i.e., when the outcomes were only measured with a single question and these cannot be measured with a scale. For comparisons of two different groups, a non-parametric Mann-Whitney-U test was used and relationships and associations between two variables have been tested with Spearman correlations. Parametric tests were used when scales have been built, e.g., by summarizing different questionnaire items into a single scale, especially the dimensions place attachment and psychological restoration where the outcome variables were measured with three questions each and the mean was calculated to have one metric value for the scale. Linear mixed models (LMM) were carried out to assess the influence of different predictor variables onto the dependent variable (outcome variable) simultaneously. All LMMs were built with place as a random effect, the variable birder vs. non-birder and the variable experience of a highlight (yes/no) as fixed

factors. Place was used as a random effect to correct statistically for the influence of the different places, and to assess whether place itself has a significant influence. As the three different outcome variables, we used psychological restoration, place attachment and visit satisfaction as dependent variable in three separate LMMs. SPSS 28 was used for all analyses.

3. Results

3.1. Comparison of birders with non-birders

We first analysed differences between birders and non-birders. There were significant differences between non-birders and birders (Table 2). Non-birders had a higher satisfaction with their visit, while birders showed a higher willingness to revisit, birders had a longer visit duration, and a higher visitation frequency. There were no differences in distance travelled, psychological restoration, place attachment and age (Table 2).

3.2. Associations of place attachment and psychological restoration

There were significant correlations with the outcome variables place attachment and psychological restoration (see Table 3). Some results showed the same direction in birders and non-birders, while others diverge. Concerning similarities, a higher visitation frequency was related to a higher place attachment, meaning that people feel more attached to a place when they visit it more often and vice versa. Further, satisfaction visit was positively correlated with place attachment and psychological restoration. People who describe their visit as more satisfying also reported a higher attachment toward this place, and also a higher psychological restoration. Willingness to revisit (loyalty) was also positively related to place attachment, i.e., people who felt more attached also were more in favor of re-visiting that place in the future.

Diverging results existed in distance travelled. In non-birders, it was positively correlated, while in birders it was negatively related to place attachment. The place loyalty (willingness to revisit) was unrelated to psychological restoration in birders, but was positively related in non-birders. Willingness to pay (entrance fee) was positively related to place attachment and psychological restoration in non-birders, but unrelated in birders (Table 3). Duration of a visit was positively correlated to place attachment in birders, but not in non-birders.

3.3. The experience of a “highlight” during the visit

Concerning psychological restoration, highlights influenced birders and non-birders simultaneously, but non-birders experienced a higher psychological restoration. In the linear mixed model (LMM), there was a significant influence of the two independent variables birder (yes/no) and experience of a highlight (yes/no) on psychological restoration

Table 2

Differences between birders and non-birders in place attachment and psychological restoration. As scale means were compared in psychological restoration and place attachment, T-tests were used. Because the other variables were non-parametric, a Mann-Whitney-U Test was applied. Table shows means and SD for birders and non-birders and T and P-value for parametric test, and Mann-Whitney-U, Z, and P value for non-parametric tests.

	Non-Birder		Birder		T-value	P
	Mean	SD	Mean	SD		
Place attachment	3.76	0.97	3.83	0.79	−0.913	0.362
Psychological restoration	4.42	0.54	4.34	0.57	1.395	0.164
Age	52.08	14.81	53.22	14.31	−0.793	0.428
					Mann-Whitney-U	Z
Visit satisfaction	4.33	0.61	3.97	0.74	16402.5	−5.097
Loyalty: willingness to revisit	4.01	1.13	4.36	0.92	18267.0	−3.275
Willingness to pay entrance fee	1.83	0.66	1.96	0.82	19558.0	−1.635
Visit duration	2.37	0.95	2.83	1.07	16500.5	−4.800
Visit frequency	2.28	1.52	2.90	1.56	16537.5	−4.612
Travel distance	3.35	1.85	3.35	1.65	21581.5	−0.182
						P
						<.001
						0.001
						0.102
						<.001
						<.001
						0.855

Table 3Correlation between the scales of place attachment and psychological restoration in birders and non-birders. ** denotes $p < 0.01$.

		Non-Birder		Birder	
		Place Attachment	Psychological restoration	Place Attachment	Psychological restoration
Visit frequency	rho	.360**	0.037	.362**	0.047
	p	<.001	0.617	<.001	0.47
	N	185	185	237	239
Travel distance	rho	−0.056	.227**	−.167*	0.003
	p	0.447	0.002	0.01	0.966
	N	184	184	235	237
Visit satisfaction	rho	.300**	.435**	.273**	.447**
	p	<.001	<.001	<.001	<.001
	N	185	186	237	239
Loyalty: willingness to revisit	rho	.466**	.215**	.364**	0.089
	p	<.001	0.003	<.001	0.168
	N	183	184	237	239
Willingness to pay entrance fee	rho	.210**	.188*	0.073	0.096
	p	0.004	0.011	0.268	0.144
	N	182	183	230	232
Visit duration	rho	0.115	0.031	.141*	0.092
	p	0.12	0.675	0.031	0.154
	N	185	186	237	239

Table 4

Linear mixed model with psychological restoration as dependent variable, and the predictors birder vs. non-birders, and the experience of a highlight. Place was used as random variable. SE = standard error, T = T value.

Source of variance	F	Sig.	Coefficient	SE	T
Constant	14726.11	<.001	4.388	0.044	99.615
Birder – Non-birder	4.269	0.040	0.061	0.065	0.941
Highlight (yes/no)	5.325	0.022	−0.214	0.090	−2.378
Birder * Highlight	1.261	0.262	0.141	0.125	1.123

(Table 4). The interaction term was not significant. This first indicates that non-birders generally showed a higher psychological restoration than birders, and second, that people experiencing a highlight significantly scored higher on psychological restoration. Both effects were in the same direction, and thus, additive (Fig. 2).

The random effect of place was not significant and accounted for 6.54% of the variance (Wald $Z = 0.320$, $p = 0.749$).

Concerning visit satisfaction as a dependent variable, non-birders reported a higher visit satisfaction, and people experiencing a highlight also reported a higher visit satisfaction (Table 5, Fig. 3).

The random effect of place was not significant and accounted for 2.75% of the variance (Wald $Z = 1.107$, $p = 0.268$).

Results concerning place attachment apparently retrieved similar, albeit non-significant results ($p > 0.05$ in both variables; Table 6). Therefore, birders and non-birders did not differ in place attachment and

Table 5

Linear mixed model with visit satisfaction as dependent variable, and the predictors birder vs. non-birders, and the experience of a highlight. Place was used as random variable. SE = standard error, T = T-value.

Source of variance	F	Sig.	Coefficient	SE	T
Constant	5134.453	<.001	4.043	0.066	61.668
Birder – Non-birder	27.825	<.001	0.295	0.082	3.622
Highlight (yes/no)	7.949	0.005	−0.341	0.111	−3.061
Birder * Highlight	2.628	0.106	0.250	0.154	1.621

the observation of a highlight had no significant effect on place attachment. The random effect of place was not significant and accounted for = 2.37% of the variance (Wald $Z = 0.807$, $p = 0.419$). Thus, place attachment was not influenced by the fact that a person is a birder or has experienced a highlight.

4. Discussion

Concerning RQ1 and RQ2, the data show that birders and non-birders differ in their travel outcomes. Although birders showed a lower satisfaction and restoration, their destination loyalty and visitation frequency was higher, and visit duration longer. At first, this seems counterintuitive, but may be related to the leisure activity of birding. As birding can be a strenuous activity, and – in terms of birding “output” – less rewarding, birders feel connected and loyal to good birding places

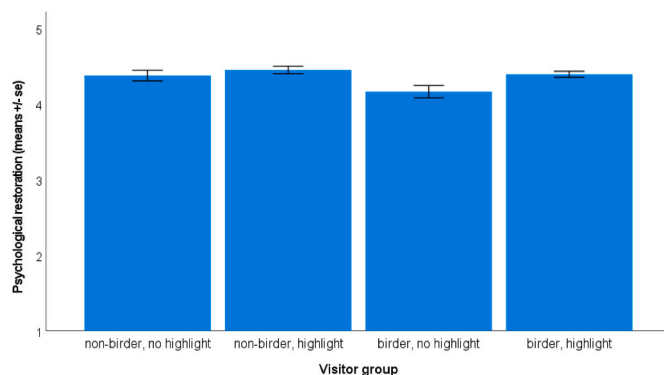


Fig. 2. Psychological restoration of visitors. Comparison of the visitor groups birder and non-birder in relation to a subjectively experienced highlight. Means ± 1 standard error.

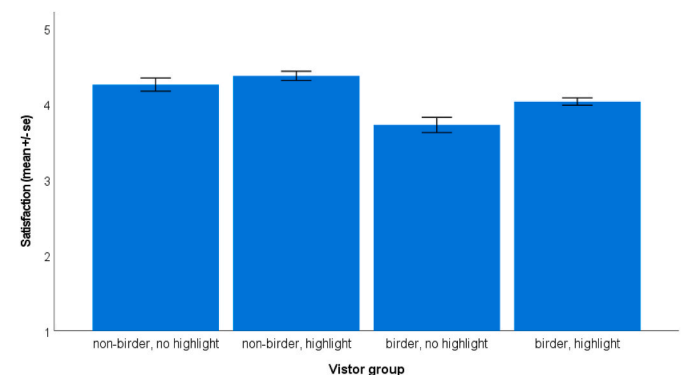


Fig. 3. Satisfaction of visitors. Comparison of the visitor groups birder and non-birder in relation to a subjectively experienced highlight. Means ± 1 standard error.

Table 6

Linear mixed model with place attachment as dependent variable, and the predictors birder vs. non-birders, and the experience of a highlight. Place was used as random variable.

Source of variance	F	Sig.	Coefficient	SE	T
Constant	2682.455	<.001	3.907	0.084	46.457
Birder – Non-birder	0.650	0.421	–0.11	0.105	–1.053
Highlight (yes/no)	3.623	0.058	–0.38	0.142	–2.681
Birder * Highlight	3.826	0.051	0.386	0.197	1.956

(which explains the willingness to revisit) even if the current visit was somewhat unsuccessful, i.e., without a “good” bird spotted, or even more with a good bird missed. Thus, birders are more goal-oriented during their leisure activity (compared to non-birders) which can explain the difference (Applegate & Clark, 1987; Mutanga et al., 2017). Concerning RQ3, the statistical correlations regarding place attachment and psychological restoration, there were differences between birders and non-birders. The destination loyalty (willingness to revisit) was unrelated to psychological restoration in birders but was positively related in non-birders. This suggest that non-birding tourists may decide about a revisit, or their loyalty based on their current experience, while birders may decide about their loyalty in terms of productivity of a given birding hotspot, i.e., decide to visit again, even when the current visit was unproductive. The decision to revisit a place might be based on a different reason rather than restoration, mainly because of the bird species diversity of the place.

Some previously published results have been confirmed in our study: Visit frequency was related to place attachment, thus, more visits produced a higher attachment, in both birders and non-birders. This fits to other research that suggests that a higher “insidedness” results in a higher attachment (see review in Lewicka, 2011). In this study, destination loyalty was related to place attachment, and place attachment can be a direct or indirect predictor of loyalty (Stylidis et al., 2020; Tasci et al., 2022). Also, place attachment led to a higher visitor satisfaction (Lee & Lee, 2017; Prayag & Ryan, 2012). Further, the length of a visit was positively related to restoration and wellbeing (Carrus et al., 2015; Nghiem et al., 2021).

Concerning RQ4, the experience of a highlight predicted psychological restoration and visit satisfaction in both birders and non-birders, while place attachment was unaffected by a highlight. Birders who experienced no highlight during their visit were less satisfied and felt less psychologically restored. This can be conceptually related to a consumptive leisure activity such as hunting or fishing, a “good bird” can have the function of a kind of trophy (Goodfellow, 2013). Thus, the visit of a good birding place has different function and meaning in birders and non-birders. Our results are supported by previous work of Applegate and Clark (1987). and may be related to the goal-sensitivity of birders. It is also in congruence with the study of Randler et al. (2023) who showed that more advanced birders preferred species that are rare, difficult to identify, and shy. So, the encounter probability for an experienced birder with a good bird may be lower compared to more casual birders or even un-experienced non-birders. The experience of a highlight may have some managerial implications (see below), but seems relevant in both birders and non-birders, because the importance of wildlife experience was also mentioned previously (Lucrezi et al., 2022; Mutanga et al., 2017). Moreover, in highly specialized birders, the enjoyment of nature as motivation for birding was negatively related to membership in an elite-birding organization (Randler, 2023). This can be interpreted in a way that non-members of this bird organization may primarily go out to watch birds, while experienced and highly skilled birders may go birding to discover “something special”, which can be experienced as some kind of goal-orientation.

Psychological restoration and visit satisfaction were significantly influenced by experiencing a highlight, which can be interpreted as an antecedent of visit satisfaction (Chi & Qu, 2008), but place attachment

was unrelated. Both, psychological restoration and visit satisfaction are conceptualized as situational assessments, place attachment may be more a trait-like measure in our study. Therefore, the current experiencing of a highlight has no influence on place attachment, and being unsatisfied with one specific visit may have no strong influence of place attachment. For future studies it might be interesting to ask for the history of bird highlights at the respective place because a history of many “good birds” may help through a lean period.

5. Limitations

The study was carried out in a quantitative manner and could have benefitted from additional, more detailed interviews. It was restricted locally to SW Germany, and could be extended to larger areas, and also to other dimensions of nature experience in addition to birdwatching. Seasonal effects have not been addressed because the study was carried out during the breeding season. Thus, winter visitors would be another interesting research study.

6. Conclusion

One conclusion is that different groups of visitors of birdwatching sites differ in psychological restoration, place attachment and visit satisfaction. This aspect is novel because it compares birders and non-birders in environments that are identical, and during the same time period. Further, the importance of a highlight experienced during the visit has emerged as an important topic that could be relevant for the managerial implications. The results are generalizable because we studied eight different birdwatching sites and controlled statistically for them (via a random variables approach), so that the effects we have found are similar across the sites.

CRedit authorship contribution statement

Christoph Randler: Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft. **Selina Friedrich:** Conceptualization, Data curation, Investigation, Methodology, Writing – review & editing, The authors declare that they have no competing interests. **Sophia Koch (née Nagel):** Data curation, Investigation, Methodology, Writing – review & editing, The authors declare that they have no competing interests.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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